ACADEMIC RESEARCH PROJECT (MBA-WILP)

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE PILANI

EC1 Experiential Learning – **Situated Learning Project**

AI Application Ecosystem (S1-25_MBACCZG580)

PHASE 2

AI Use Case Design, Feasibility, Ecosystem Mapping & Risk Analysis

Submitted by:

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Executive Summary

This Phase-2 submission proposes an AI-enabled orchestration layer to streamline the RFQ (Request for Quotation) response process used in automotive engineering services. Today, RFQ coordination relies heavily on manual email exchanges, unstructured documents, and inconsistent SME (Subject Matter Expert) routing. This results in delays, rework loops, unclear ownership, and estimation risks—especially for multi-site XiL (MIL, SIL, HIL, VIL) programs.

The proposed solution introduces two complementary AI pathways inside a unified collaboration workspace: (1) **Document Intelligence + SME Routing**, and (2) **AI-Assisted Proposal Drafting**. The system automates intake, extracts structured requirements, identifies missing sections, flags anomalies in cost sheets, proposes timelines, detects dependencies, and generates first-draft narrative sections for human approval.

Business value is expected in the form of improved turnaround time, better first-time-right quality, clear accountability, and higher proposal consistency. This aligns with Phase-2 requirements to articulate an AI use case, workflow, feasibility, ecosystem needs, risks, and benefit–trade-offs.

Context

Phase-1 identified several operational pain points: manual parsing of RFQ documents, inconsistent SME assignment, fragmented communication, unclear stage ownership, and high dependency on individual effort. These issues become more severe when proposals span multiple XiL activities.

Example Scenario Chosen for Phase-2 (Illustrative Only)

To maintain continuity with Phase-1 while offering a realistic and structured example for Phase-2, I am using a **multi-stage engineering workflow**—represented through XiL stages (MIL, SIL, HIL, VIL)—purely as an **illustrative scenario**.

They simply provide a concrete context to demonstrate how AI-supported routing, extraction, dependency identification, and drafting can improve complex RFQ processes.

This submission transforms those observations into a feasible, ethically safe, and workflow-integrated AI solution, as required by the Phase-2 rubric.

Acknowledgement

I would like to express my gratitude to my faculty mentor and academic advisors at BITS Pilani for their guidance throughout this assignment. This work has been completed independently as part of the MBA-WILP program requirements.

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Abbreviation Table

Abbreviation	Full Form
RFQ	Request for Quotation
OEM	Original Equipment Manufacturer
CFT	Cross-Functional Team
SME	Subject Matter Expert
CM%	Contribution Margin Percentage
TTL	Tata Technologies Limited
AI	Artificial Intelligence
RASIC	Responsible, Accountable, Support, Informed, Consulted
HW-SW	Hardware-Software
JSON	JavaScript Object Notation
CSV	Comma-Separated Values
PDF	Portable Document Format
KPI	Key Performance Indicator

1. Proposed AI Use Case

Title: AI-Enabled RFQ Orchestration for XiL (MIL/SIL/HIL/VIL) with Dual-Option Assistance

Option 1 — Document Intelligence + SME Routing

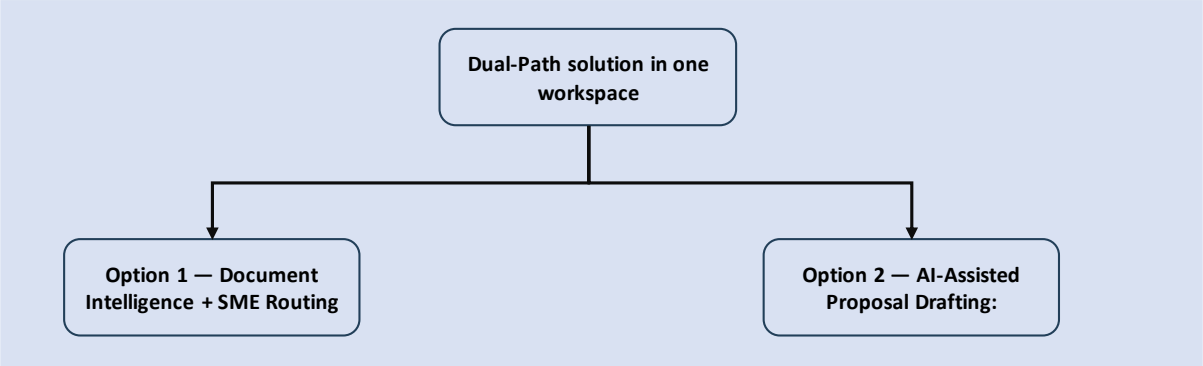
- Automated triage and classification of RFQ documents (PDF/Word, multilingual).
- Extraction of scope, deliverables, acceptance criteria, timelines, and OEM-specific template structures. Identification of missing sections, inconsistencies, and language translation needs (e.g., German → English).
- Skill-based SME routing with autogenerated subtasks, dependencies, and proposed due dates.
- Anomaly detection for large Excel cost sheets (formula mismatches, currency errors).

Option 2 — AI-Assisted Proposal Drafting

- First-draft generation of proposal sections (technical approach, assumptions, exclusions, compliance matrix).
- Context-aware phrasing using internal approved exemplars.
- Versioning and traceability for SME and EM edits.

Users of the AI Output

- Engagement Manager, SMEs (MIL, SIL, HIL, VIL), Finance, Procurement, and Delivery Leadership.



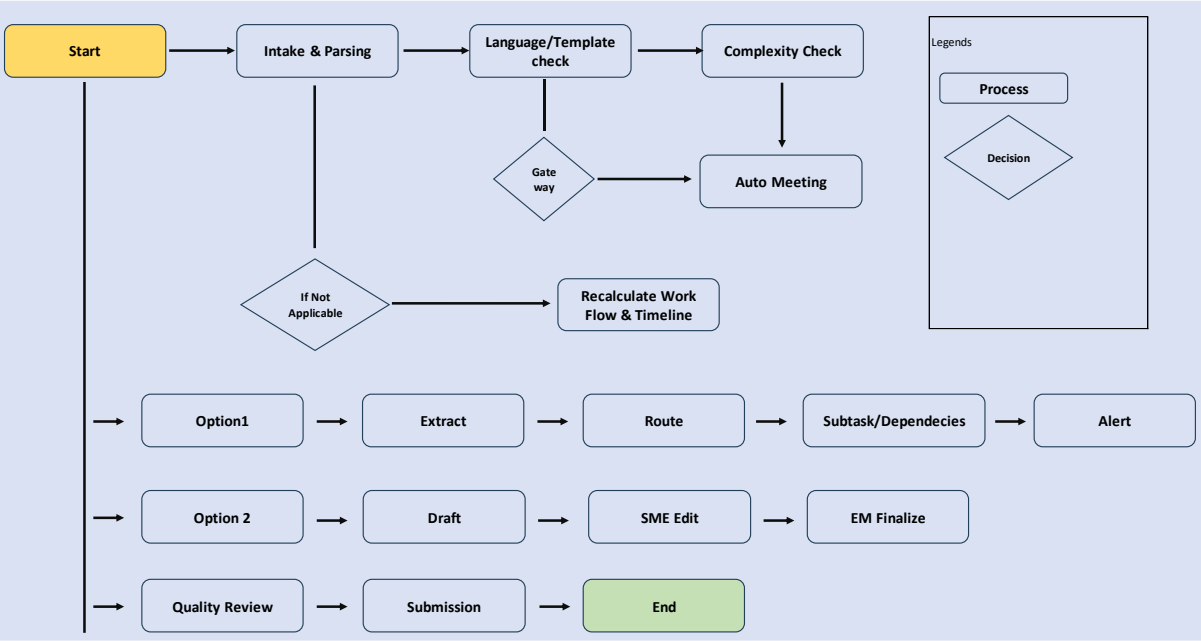
Decisions or Actions Supported

- Clarifying scope, resolving template/translation issues.
- Assigning SMEs, balancing workload, planning subtasks.
- Detecting risk items and anomalies.
- Improving narrative consistency and shortening drafting cycles.

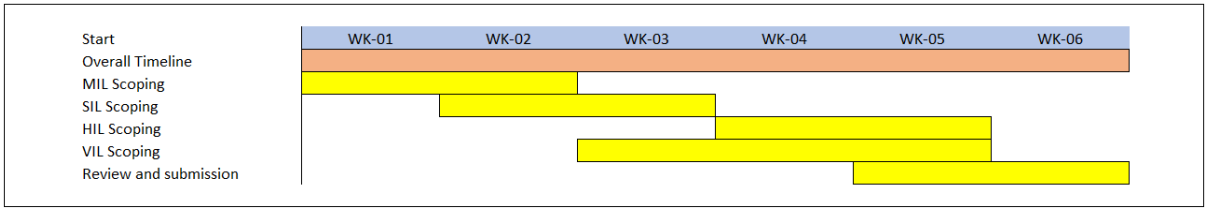
Comparison Path 1 and Path 2

Aspect	Option 1: Doc Intelligence + Routing	Option 2: AI Drafting
Main value	Clear requirements & right SME at right time	Faster writing & consistent structure
Biggest risk	Extraction errors in complex tables	Generic/off-tone text
Human role	Validate extraction & dependencies	Edit & approve draft content
Best fit	Very complex, multi-site RFQs	Repetitive sections & boilerplate

RFQ Process Workflow



Timeline dashboard



2. Target Business Outcomes

KPIs

- 30–40% faster RFQ turnaround, especially for multi-site XiL proposals.
- Higher first-time-right quality, reducing clarifications and rework.
- Clear accountability through role-based routing and automated alerts.
- Improved estimation accuracy via anomaly detection in complex cost sheets.
- Consistent proposal structure and language, increasing customer trust.
- Higher win probability through improved quality, speed, and completeness.

How AI-supported decisions will alter the workflow.

"AI-supported decisions alter the workflow by shifting manual extraction, routing, dependency identification, and drafting into automated stages, while humans move into a supervisory approval role. This reduces delays, clarifies ownership, and recalibrates timelines dynamically."

3. Detailed Workflow of the AI Solution (Non-Technical)

Inputs

- RFQ packages (PDF/Word, excel, German/English), annexures, cost sheets.
- Emails and metadata that trigger intake events.
- Past proposals, SME skills registry, dependencies from previous bids.

AI Actions

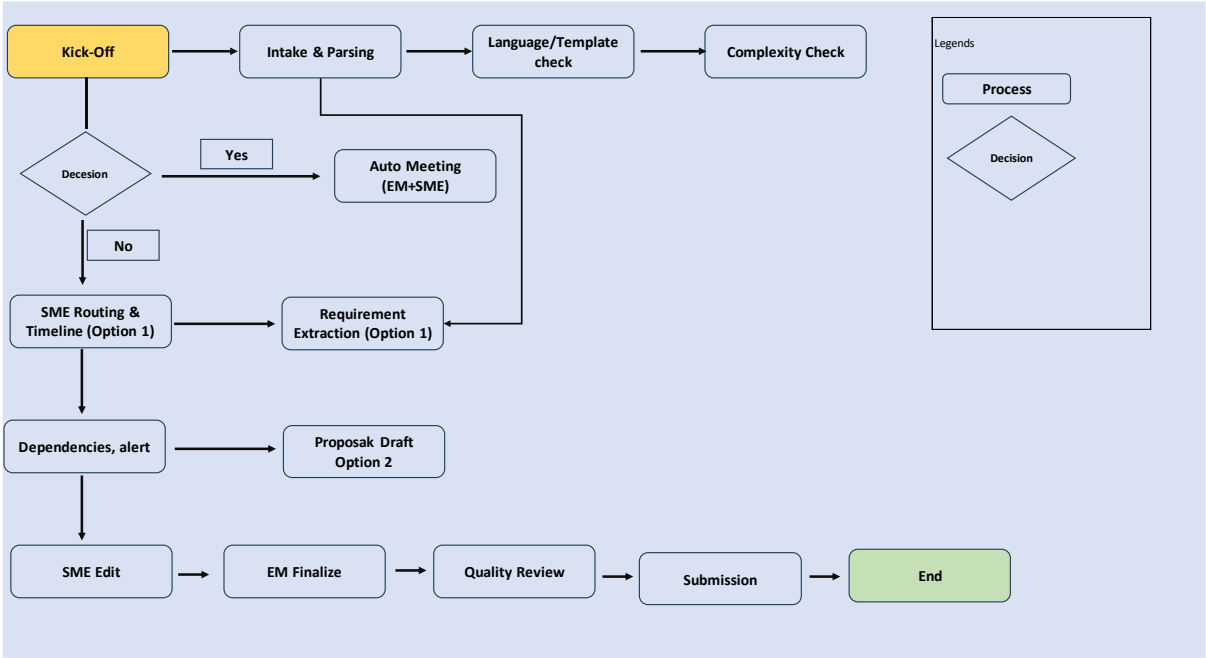
- Classify documents and map them to internal template sections.
- Identify languages and provide side-by-side translations.
- Extract XiL scopes (MIL/SIL/HIL/VIL) and related deliverables.
- Propose SME routing, subtasks, dependencies, and timelines.
- Detect anomalies in Excel cost sheets and flag them for Finance.
- Draft narrative text for proposal sections.

Outputs / Decision-Support

- A dynamic workflow board with stages, owners, due dates, and risk flags.
- First-draft proposal text for human approval.
- Timeline recalculation whenever stages are edited or marked NA/Delegated.
- Alerts for delays, risks, and dependencies requiring EM intervention.

Human Interaction (Human-in-the-Loop)

- SMEs and EM review extractions, override AI suggestions, edit drafts, or mark tasks NA.
- Tagging a stage as Critical increases alert sensitivity and locks timeline windows.
- Final submission remains fully human-approved.



Detailed To-Be Workflow (AI-Enabled)

4. Data Requirements & Feasibility

Data Sources Needed

- RFQ documents (PDF/Word), annexures, multilingual content.
- Large Excel cost sheets (multi-skill, multi-site).
- SME skills registry: discipline, location, availability.
- Past proposals (cleaned and redacted for confidentiality).
- Communication metadata for alert logic.

New Data Capture Mechanisms Required

- Stage-tagging (Critical, NA, Delegated, Highlight).
- Decision logs (who approved, what changed, and when).
- Translation confirmations for key clauses.
- Anomaly notes for Excel flags.
- SME availability snapshots.

Data Issues Affecting Accuracy/Fairness

- Heavy template variance across OEMs.
- OCR extraction errors.
- Ambiguous language or incomplete RFQ inputs.
- Sparse history for niche XiL programs.

Feasibility Assessment

Technical Feasibility

- SharePoint + Outlook already support workflow triggers.
- Document intelligence (OCR, translation) and LLM drafting can run in enterprise-secured environments.

Operational Feasibility

- Adds clear structure to intake and routing.
- SMEs require modest training but no system replacement.
- RASIC clarity reduces meeting load.

Organizational Feasibility

- Minimal IT disruption; existing systems remain unchanged.
- Requires change management and governance for auditability.

Economic Feasibility

- License and integration costs modest relative to high-value RFQs.
- Pilot approach minimizes upfront risk.

Input data type matrix

Data Type	Examples	Purpose
RFQ Docs	PDF/Word (German), annexures	Requirement extraction
Cost Sheets	Large Excel (multi-site, multi-skill)	Estimation & anomaly checks
SME Skills	Discipline, site, availability	Routing & load balance
Past Proposals	Curated internal, redacted	Safe drafting patterns
Communications	Email metadata, comments	Alerts & escalation history

5. Tools, Platforms & Ecosystem Considerations

APIs / Integrations Required

- SharePoint (file access, permissions).
- Email/Calendar (alerts, auto-meetings).
- Low-code workflow engines (routing, dependency management).
- Enterprise AI services (document intelligence, translation, LLM drafting).

Existing Organizational Systems to Integrate

- File repositories and email.
- SME skills records.
- Proposal templates and standard wording libraries.

Constraints

- NDA compliance and IT security approvals.
- Budgeting for licenses and SME time.
- Data governance: role-based access, audit logs, version control.

6. Risks, Biases & Ethical Considerations

Where Errors or Unfair Decisions May Occur

- Incorrect translation of technical requirements.
- Mis-extraction of complex tables.
- Biased SME routing repeatedly assigning work to certain individuals.
- Timeline suggestions ignoring site-specific constraints.

Ethical Concerns

- Over-reliance on AI-drafted text weakening human judgment.
- Hidden model biases affecting tone or interpretation.
- Lack of explainability in routing or anomaly logic.

Guardrails Needed

- Mandatory human approvals for all critical stages.
- Explainability logs ("why this SME", "why this timeline").
- Load-balancing dashboards to avoid unfair routing.
- Time-zone-aware scheduling logic.
- Enterprise-only AI models with full audit trails.

7. Benefit–Trade-Off Assessment

Expected Benefits

- Faster and more consistent RFQ processing.
- Clear visibility into dependencies and risks.
- Improved estimation quality and fewer customer clarifications.
- Standardized proposal language and structure.
- Reduced meeting load due to automated flags and routing.

Trade-Offs and Unintended Consequences

- Adoption curve and change-resistance.
- Risk of generic AI-generated content if not reviewed.
- False positives in anomaly detection.
- Temporary productivity dip during transition.

Conclusion

The proposed AI solution introduces a practical, non-disruptive orchestration layer that strengthens RFQ processing for XiL programs without replacing existing enterprise tools. By combining document intelligence, SME routing, anomaly detection, and AI-assisted drafting—all within a human-approved framework—the approach targets measurable improvements in turnaround time, quality, and transparency. A pilot deployment using secure enterprise AI services and clear governance provides a low-risk path to validate outcomes before wider adoption.

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8. Academic Declarations

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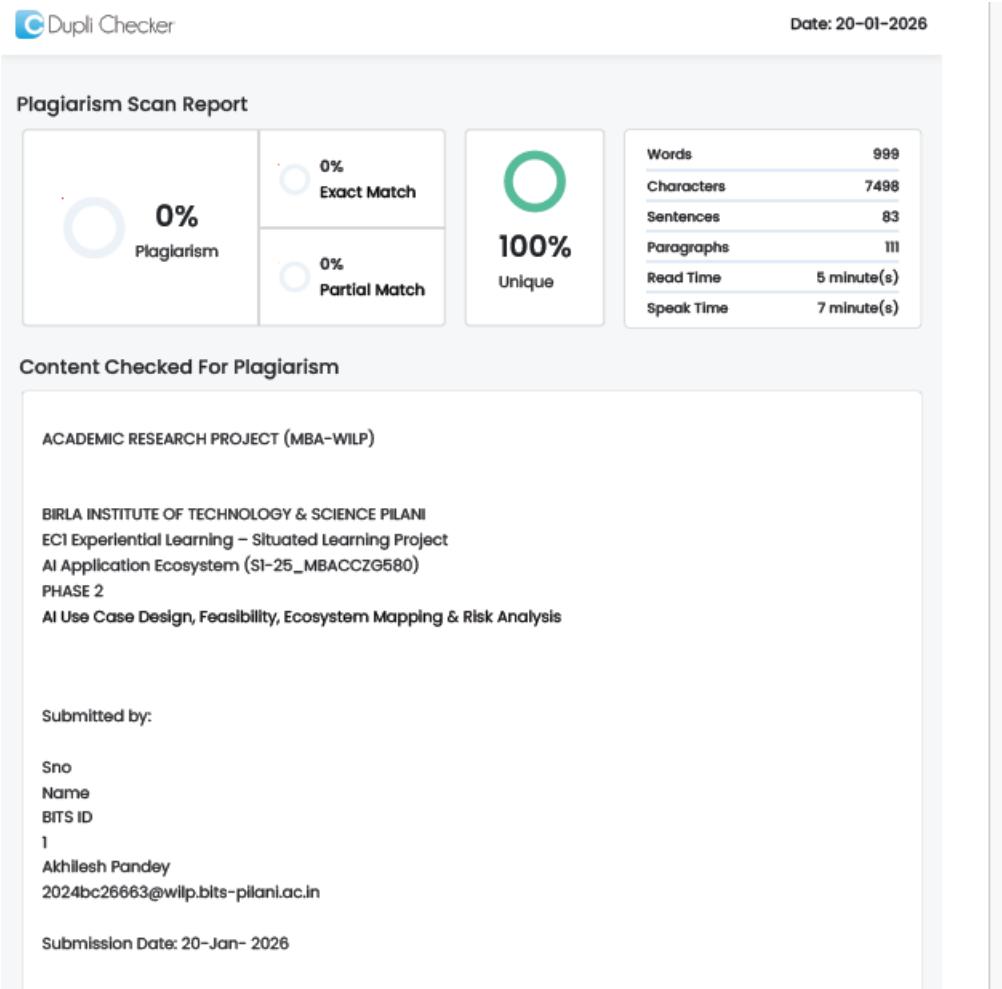

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